

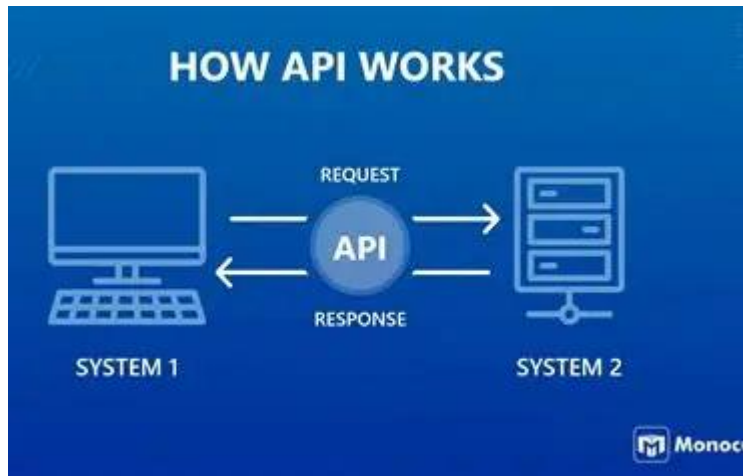
APIs, Fetch, and HTTP Requests

Research + Visual Project

Joshua Xie

This project explains:

- What APIs are
- What JSON is
- How Fetch works
- GET, POST, and PUT requests



API



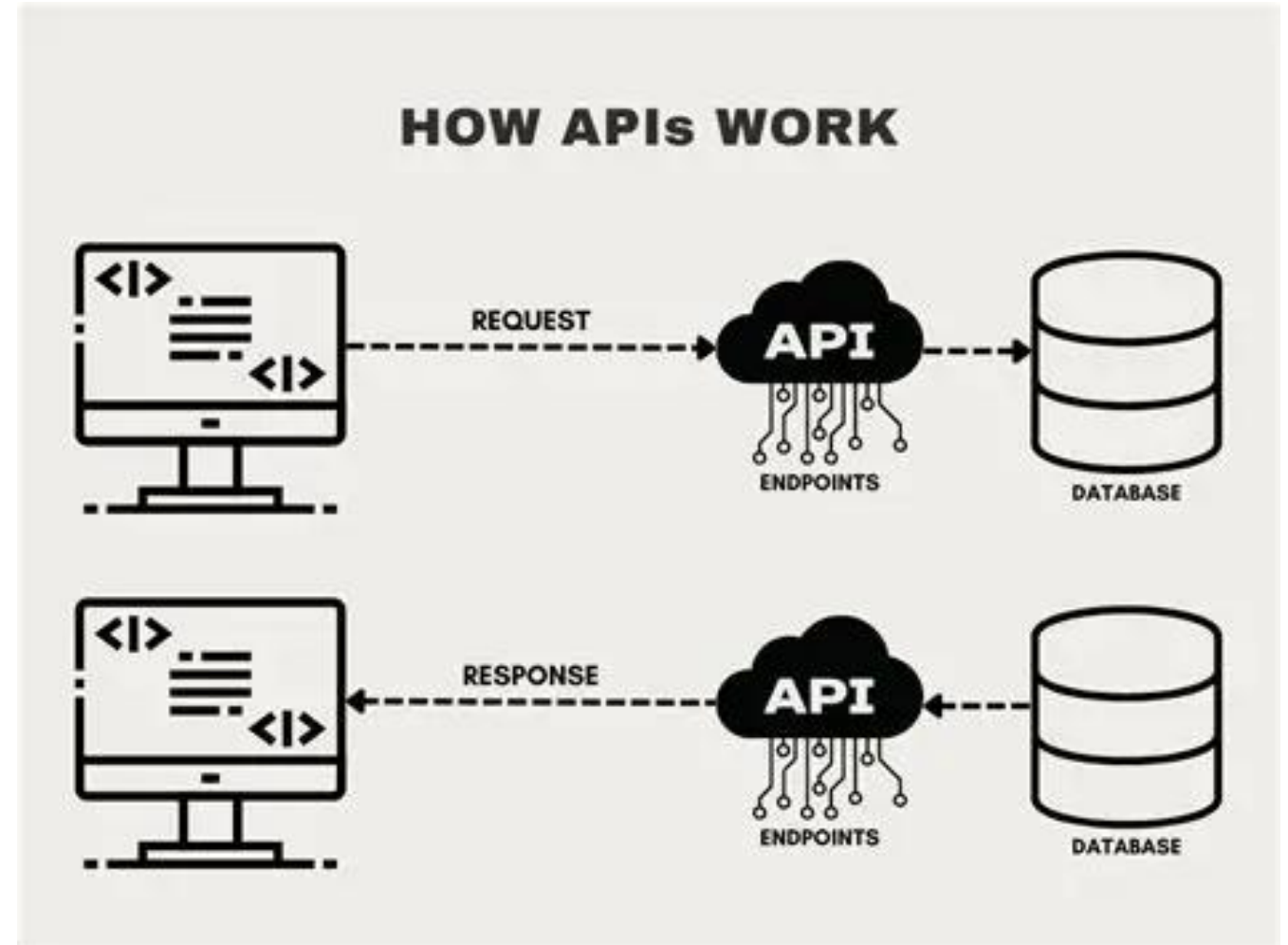
JSON



Fetch

What is an API?

- An API lets two programs talk to each other.
- One program asks for data.
- The other sends it back.
- Example:
 - A weather app asks for the weather.
 - The server sends the result.
- My understanding:
 - API is like a bridge.



API helps systems talk and share data

What is JSON?

- JSON is a simple way to store and send data.
- It is just text, not code.
- Data is written in key and value pairs.
- Example:
 - name: "Joshua"
 - age: 20
- My understanding:
 - JSON is just data.

```
{  
  "name": "Joshua",  
  "age": 20,  
  "course": "IT Programming"  
}
```

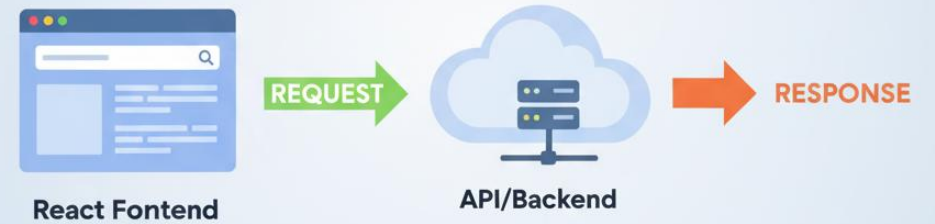
JSON shows data in a clear structure

What is Fetch API?

- Fetch is used to send requests to a server.
- It can get data or send data.
- It works with APIs.
- Example:
 - A website gets data from a server.
- My understanding:
 - Fetch is how JavaScript talks to a server.

FETCHING APIS: The Ultimate Guide

Unlock Web Data with JavaScript & React



What is a GET request?

- GET is used to get data from a server.
- It does not change data.
- Example: Get a list of users from a server.
- My understanding:
- GET is used to read data.

fetch(...) sends a request to the server

response.json() reads the JSON data

console.log(data) shows the result



```
fetch("https://example.com/data")  
  .then(response => response.json())  
  .then(data => console.log(data));
```

What is a POST request?

- POST is used to send data to a server.
- It is used to create new data.
- Example: Send a new user to the server.
- My understanding:
- POST is used to send new data.

```
fetch("https://example.com/data", {  
  method: "POST",  
  headers: {  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify({ name: "Joshua" })  
});
```

method: "POST" means sending data

headers tell the server the data is JSON

body contains the data being sent

JSON.stringify() turns data into JSON text

What is a PUT request?

- PUT is used to update data on a server.
- It replaces existing data.
- Example: Update a user's name.
- My understanding:
- PUT is used to change data.

```
fetch("https://example.com/data/1", {  
  method: "PUT",  
  headers: {  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify({ name: "New Name" })  
});
```

method: "PUT" means updating data

headers tell the server the data is JSON

body contains the new data

JSON.stringify() turns data into JSON text

GET, POST, and PUT

- GET is used to get data.
- POST is used to send new data.
- PUT is used to update data.
- My understanding:
- GET reads, POST creates, and PUT updates.

Request	What it does	Simple
GET	Gets data	Read
POST	Sends new data	Create
PUT	Updates data	change

These requests help websites communicate with servers.

References

- MDN Web Docs – API, JSON, Fetch, HTTP Methods
W3Schools – Fetch API and HTTP Requests

Thank you!